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Hoffman et al.

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(54) **INFANT ENTERTAINMENT DEVICE**

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A63H 33/00 (2006.01)

(52) **U.S. Cl.**
CPC *A63H 33/006* (2013.01)

(58) **Field of Classification Search**
CPC A63H 33/006
USPC 446/227
See application file for complete search history.

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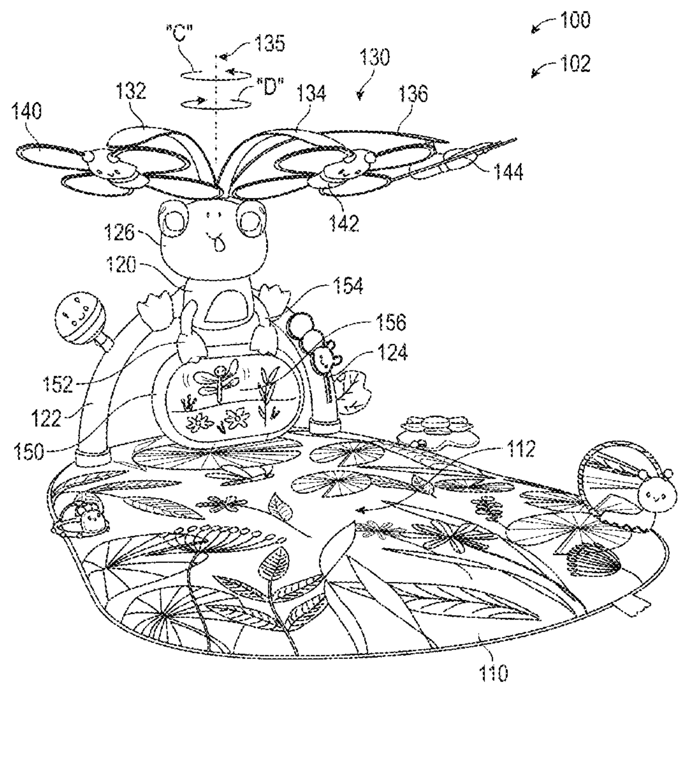
(57) **ABSTRACT**

An aspect of the infant entertainment device relates to an entertainment component that is moved by an input from an infant. In one embodiment, the entertainment component moves in response to a kicking input from an infant in a supine position. The infant entertainment device includes an input mechanism that is positioned so that the infant can move the input mechanism by kicking it. The infant entertainment device includes a drive mechanism that converts movement of the input mechanism into motion of the entertainment component. In one implementation, the entertainment component is a mobile that moves when the infant kicks the input mechanism.

17 Claims, 16 Drawing Sheets



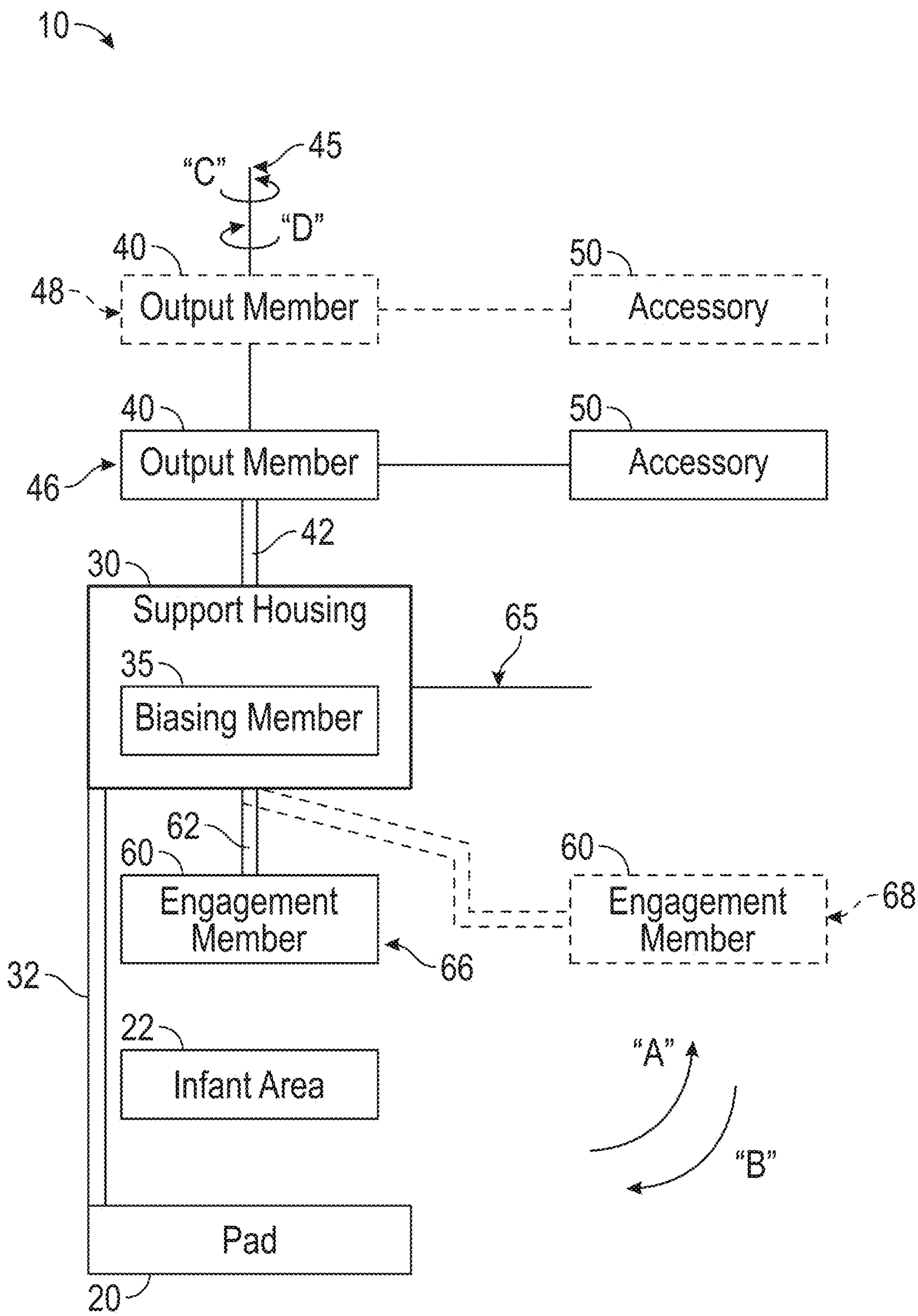

10
45
"C"
"D"
40
50
48
Output Member
Accessory
40
50
46
Output Member
Accessory
42
30
Support Housing
65
35
Biasing Member
62
60
60
Engagement Member
Engagement Member
68
66
32
22
Infant Area
"A"
"B"
Pad
20

FIG. 1

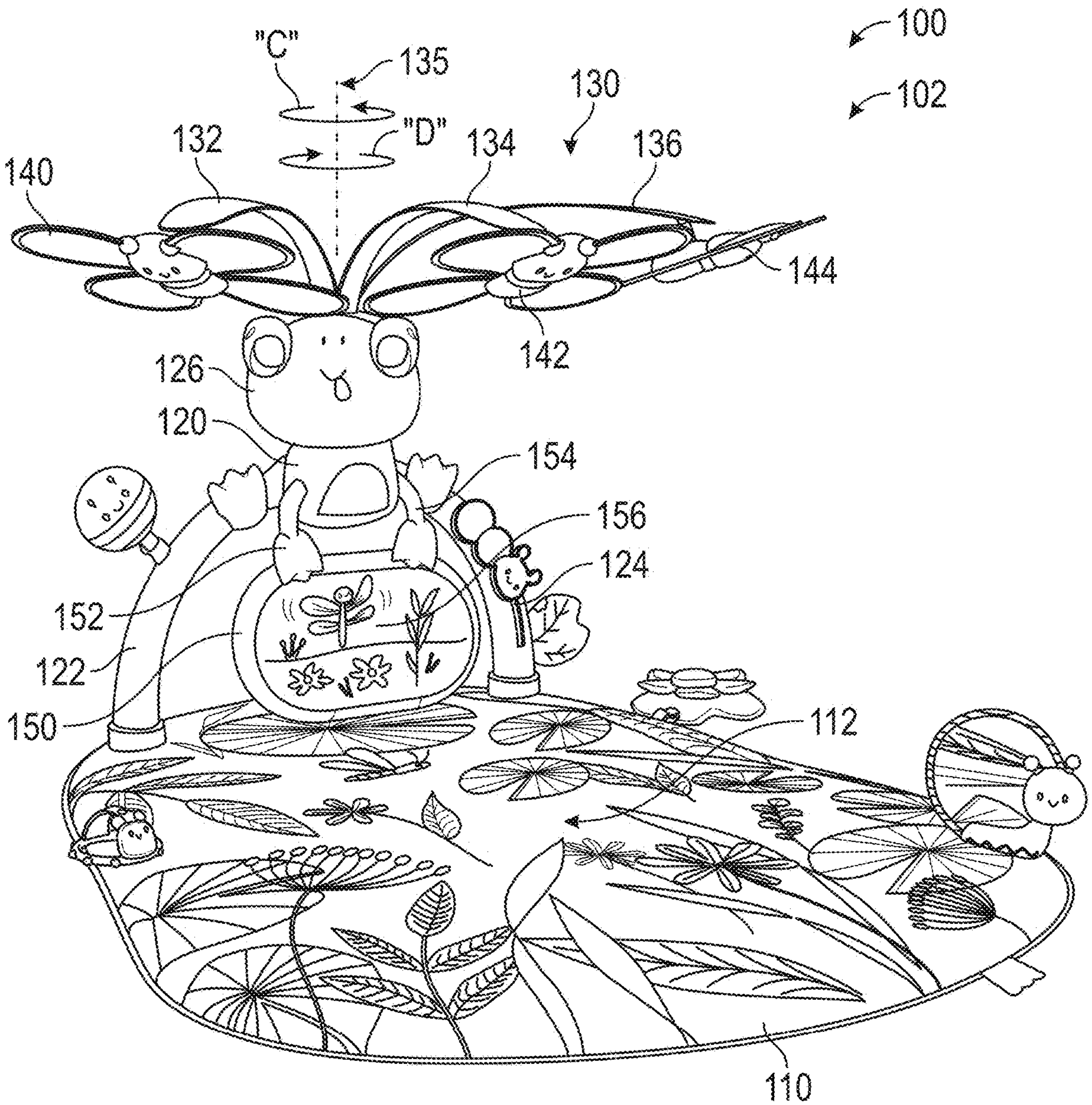

100
102
"C"
135
"D"
130
132
134
136
140
144
142
126
120
154
156
124
152
122
150
112
110

FIG. 2

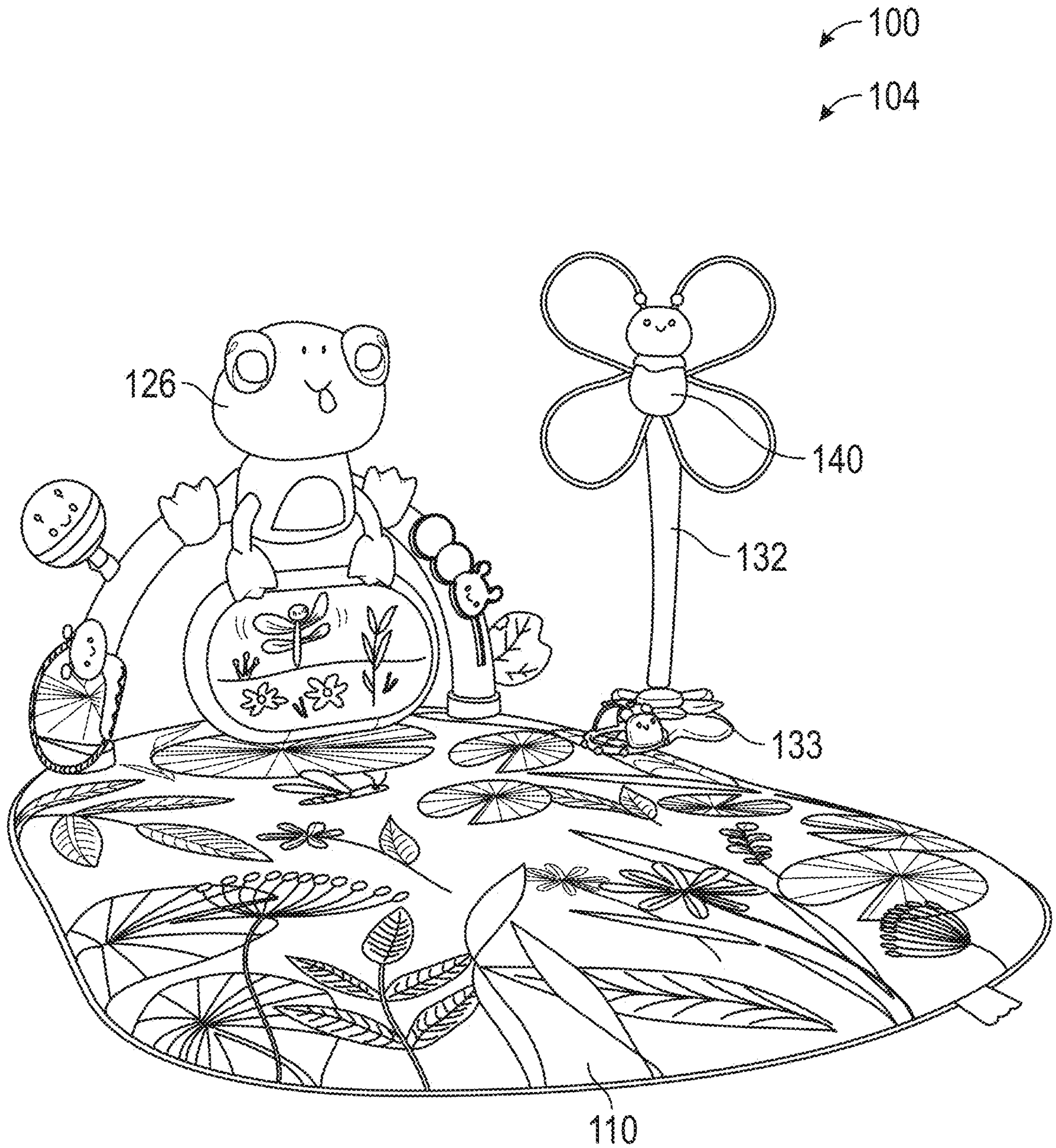


FIG. 3

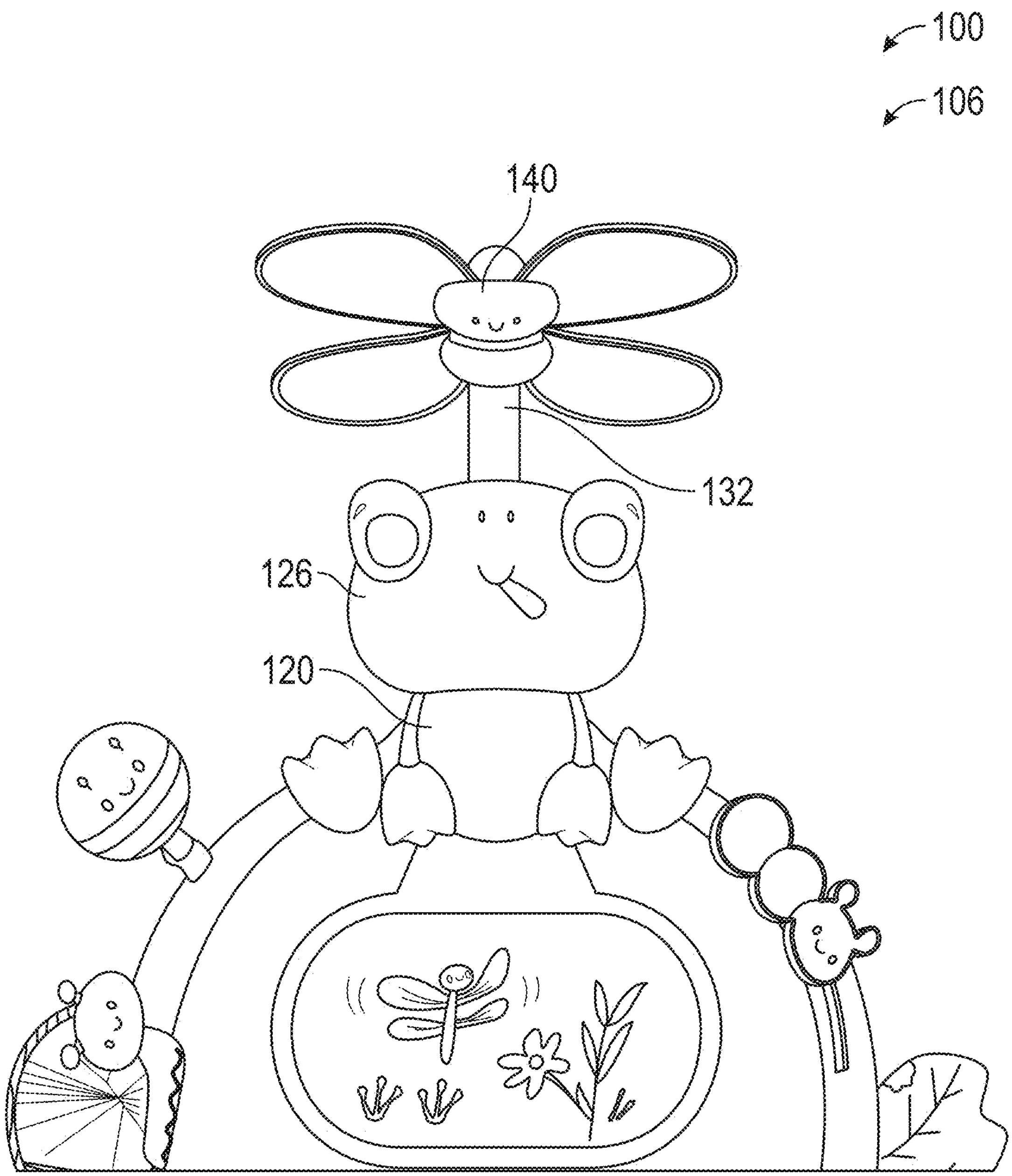


FIG. 4

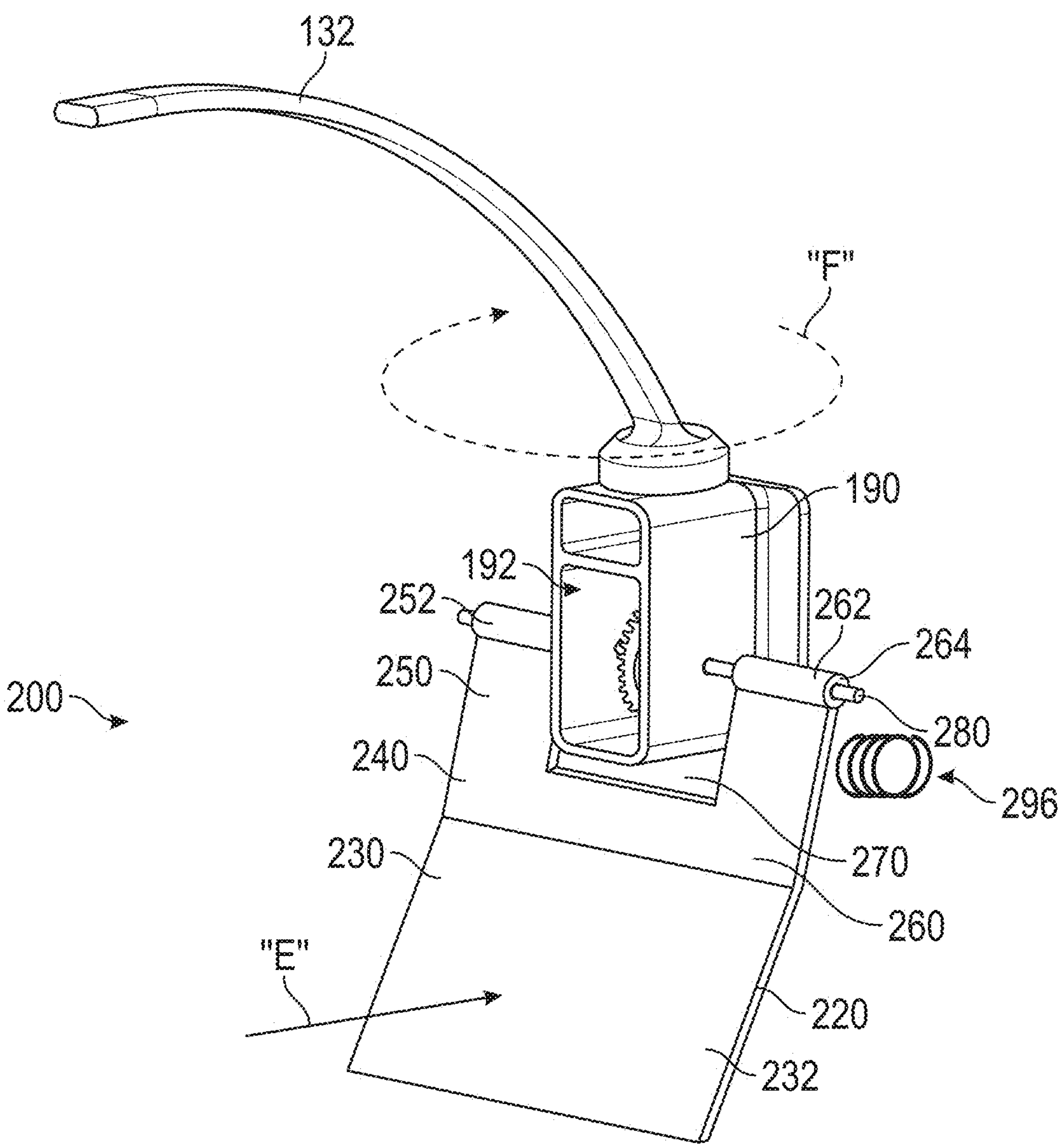

132
"F"
190
192
252
262
264
250
200
280
240
296
270
230
260
"E"
220
232

FIG. 5

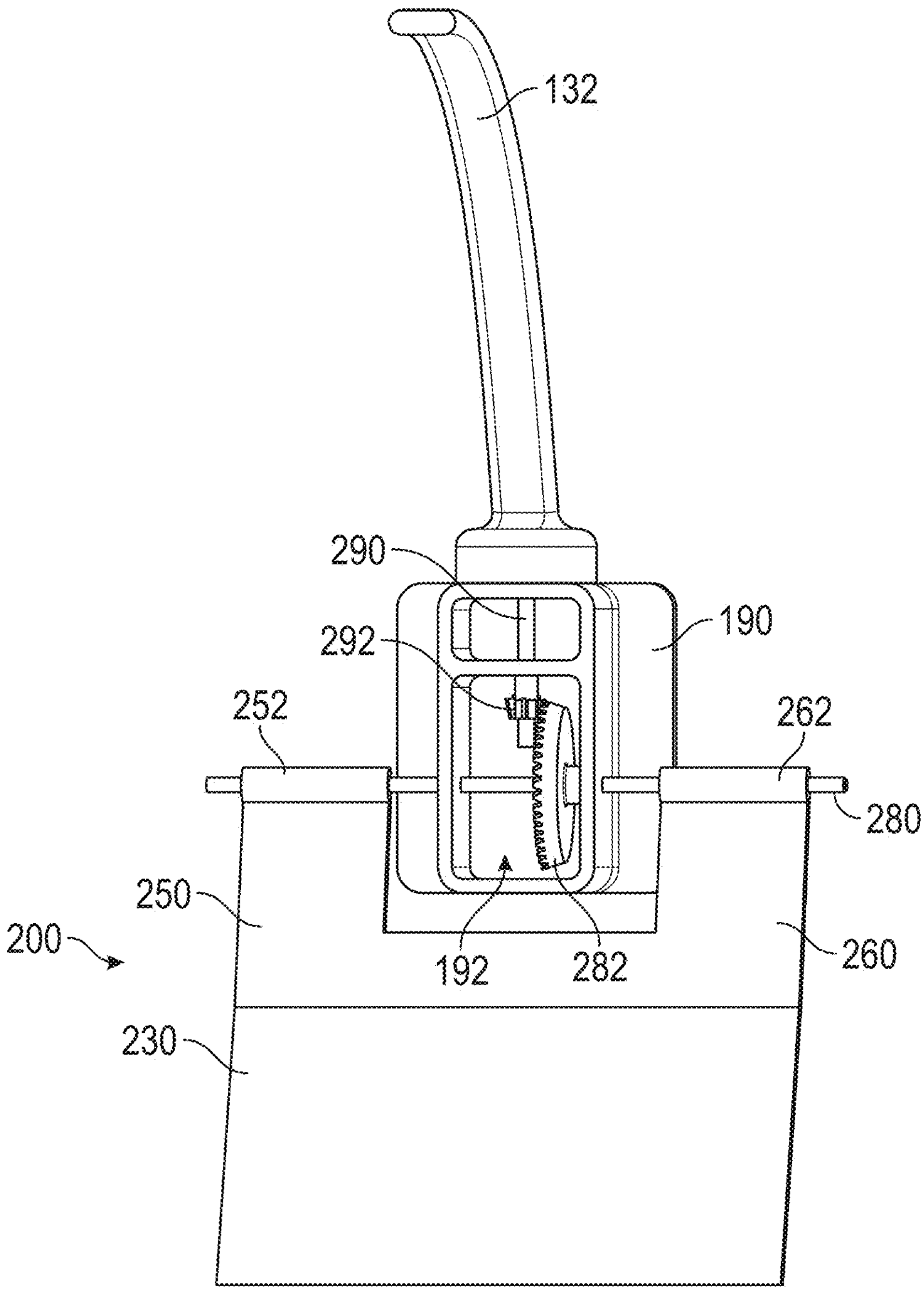


FIG. 6

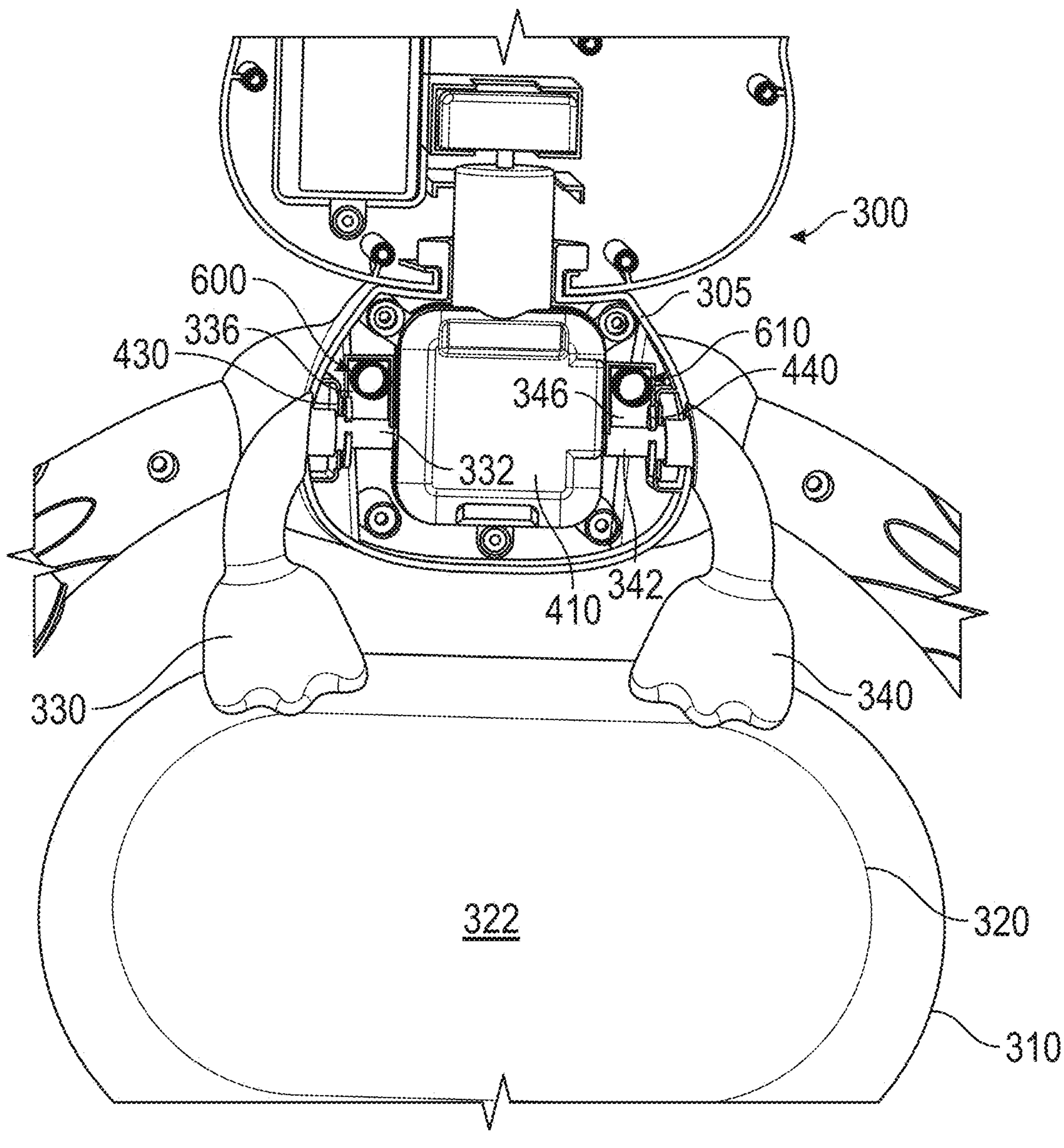


FIG. 7

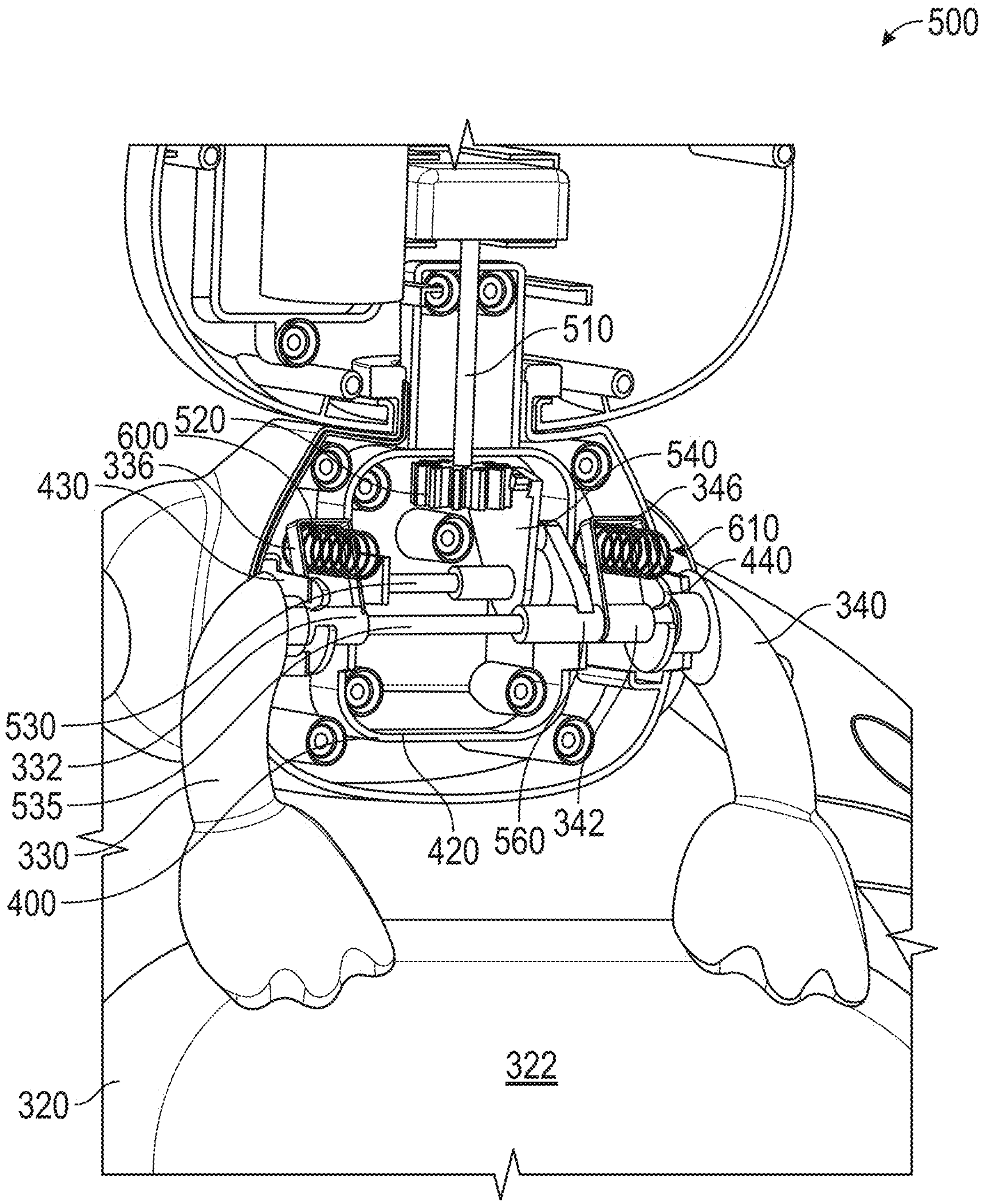

500
510
520
600
336
430
540
346
610
440
340
530
332
535
330
400
420
560
342
322
320

FIG. 8

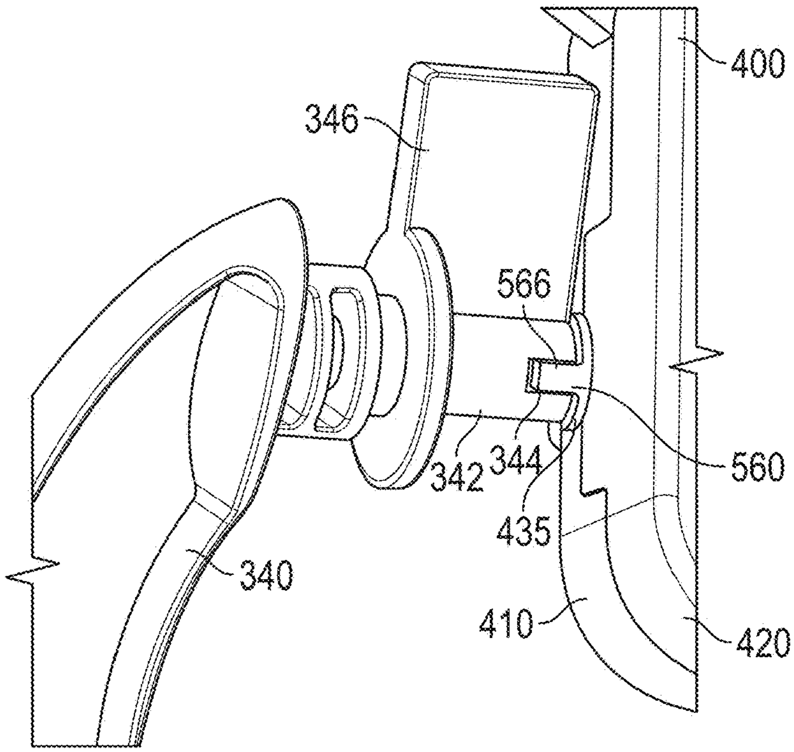
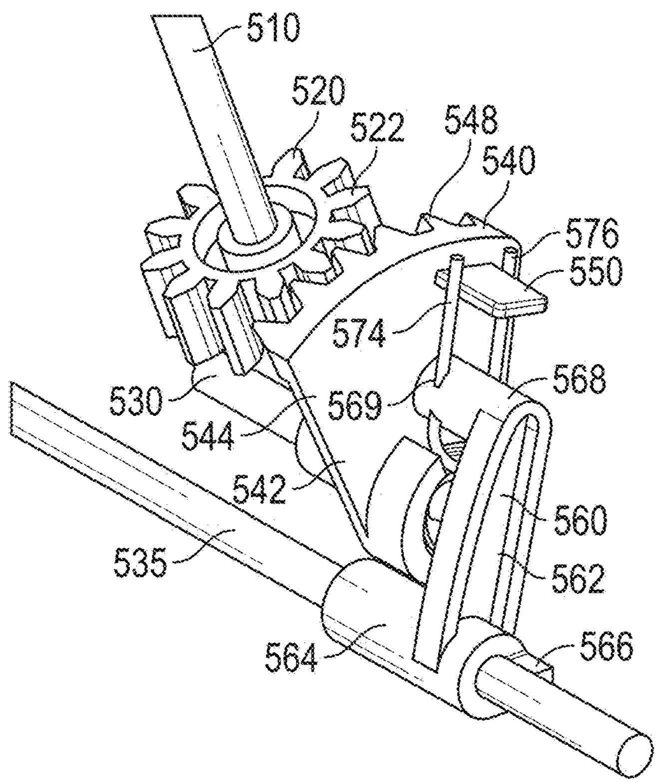

400
346
566
560
342
344
435
340
410
420

FIG. 9


510
520
522
548
540
576
550
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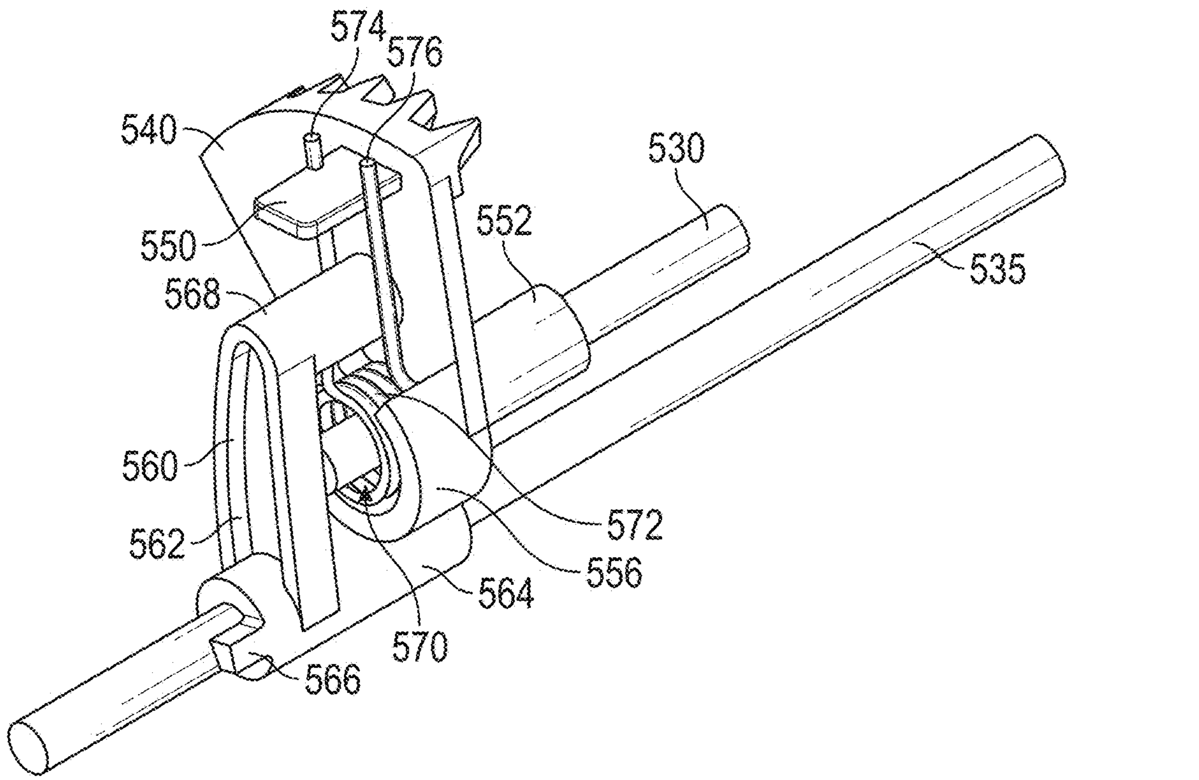
530
569
568
544
542
560
535
562
564
566

FIG. 10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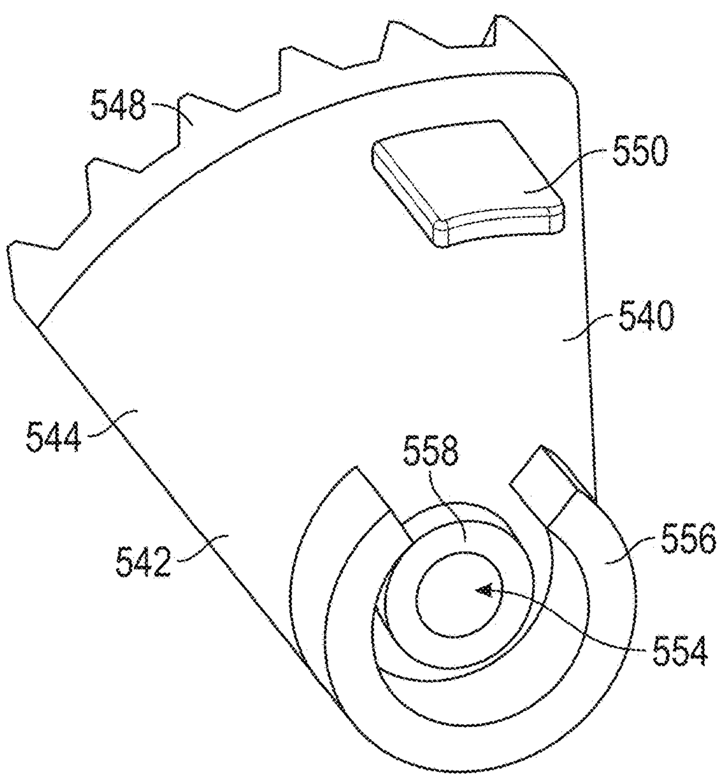
564
570
566

FIG. 11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556
554

FIG. 12

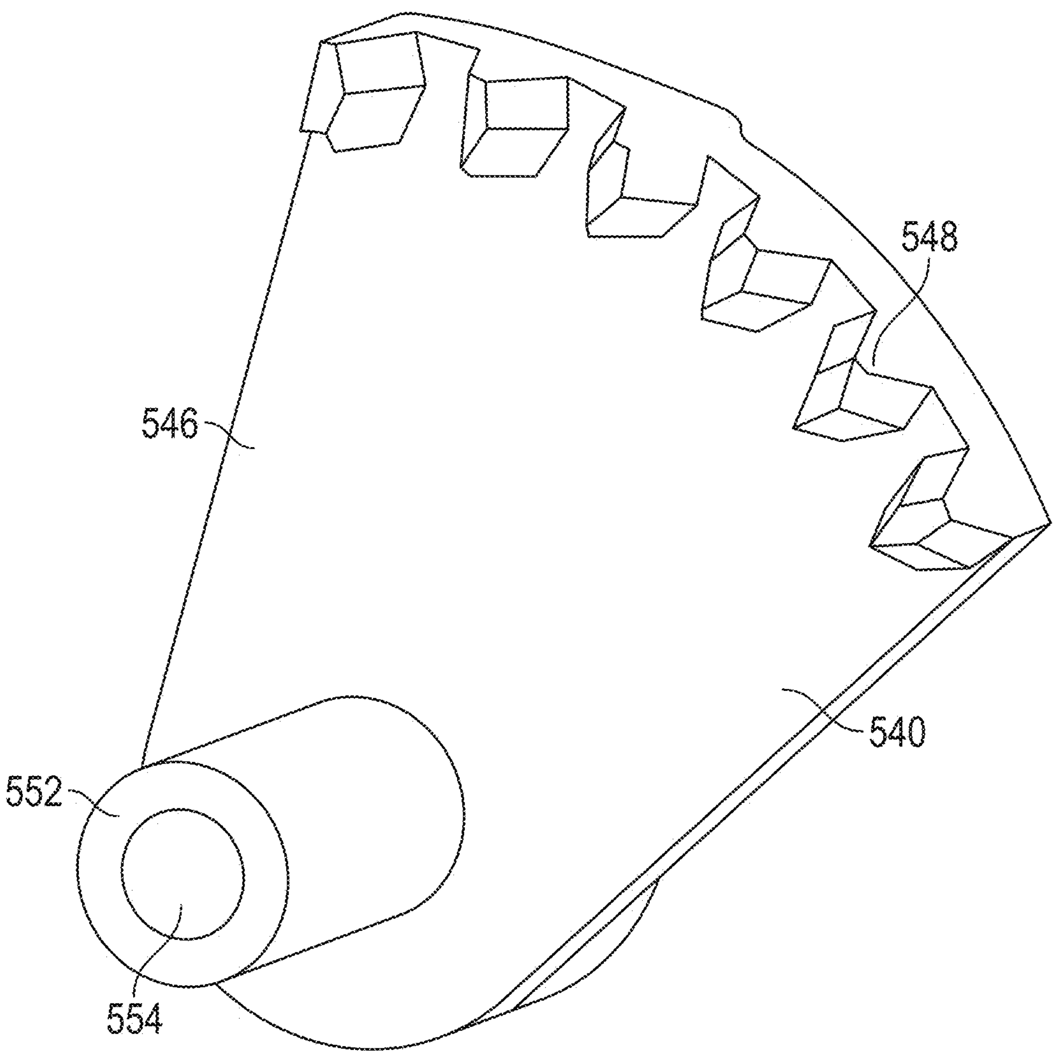

FIG. 13

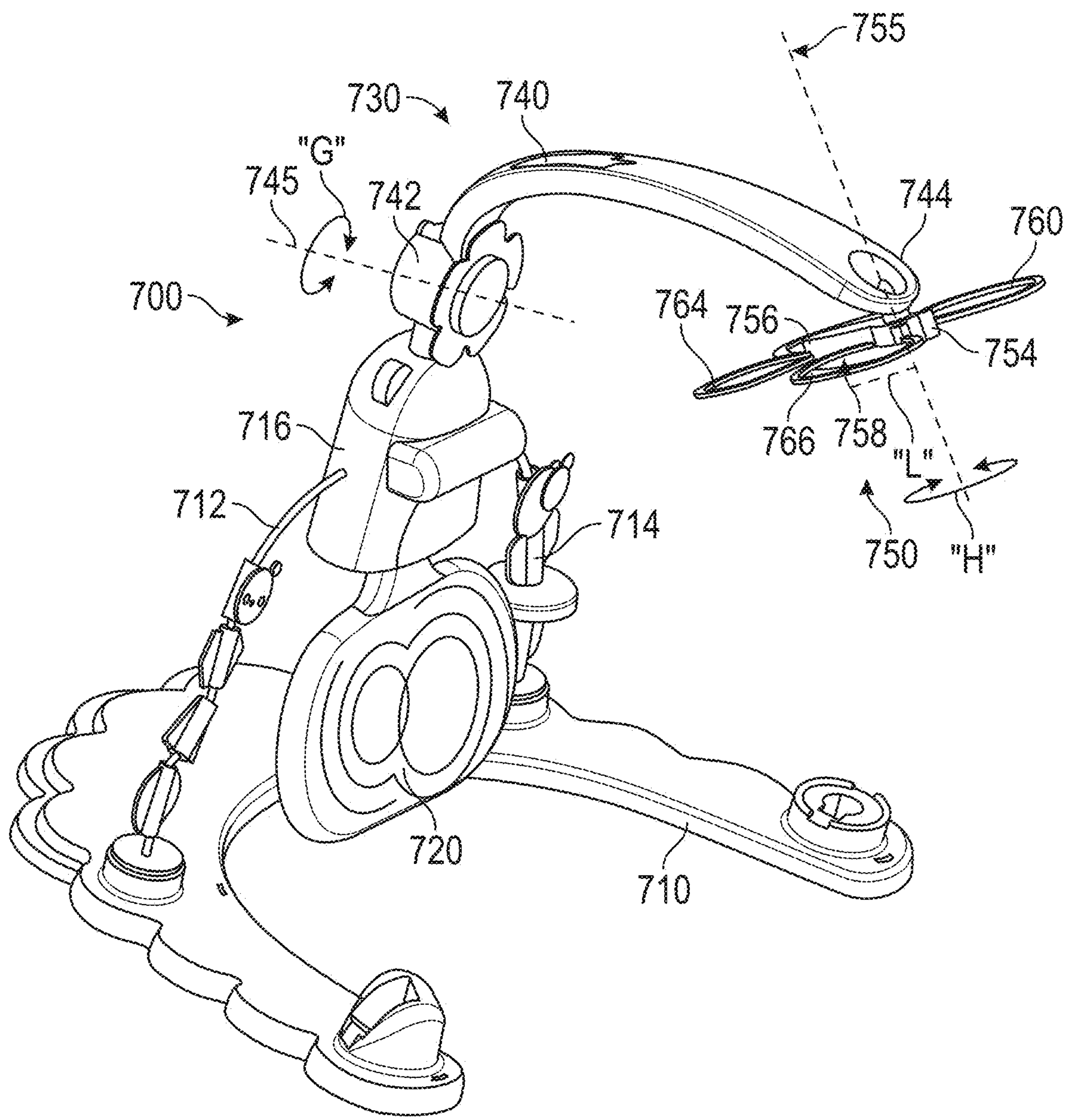
755
730
740
"G"
745
742
744
760
700
764
756
754
766
758
716
"L"
712
714
750
"H"
720
710

FIG. 14

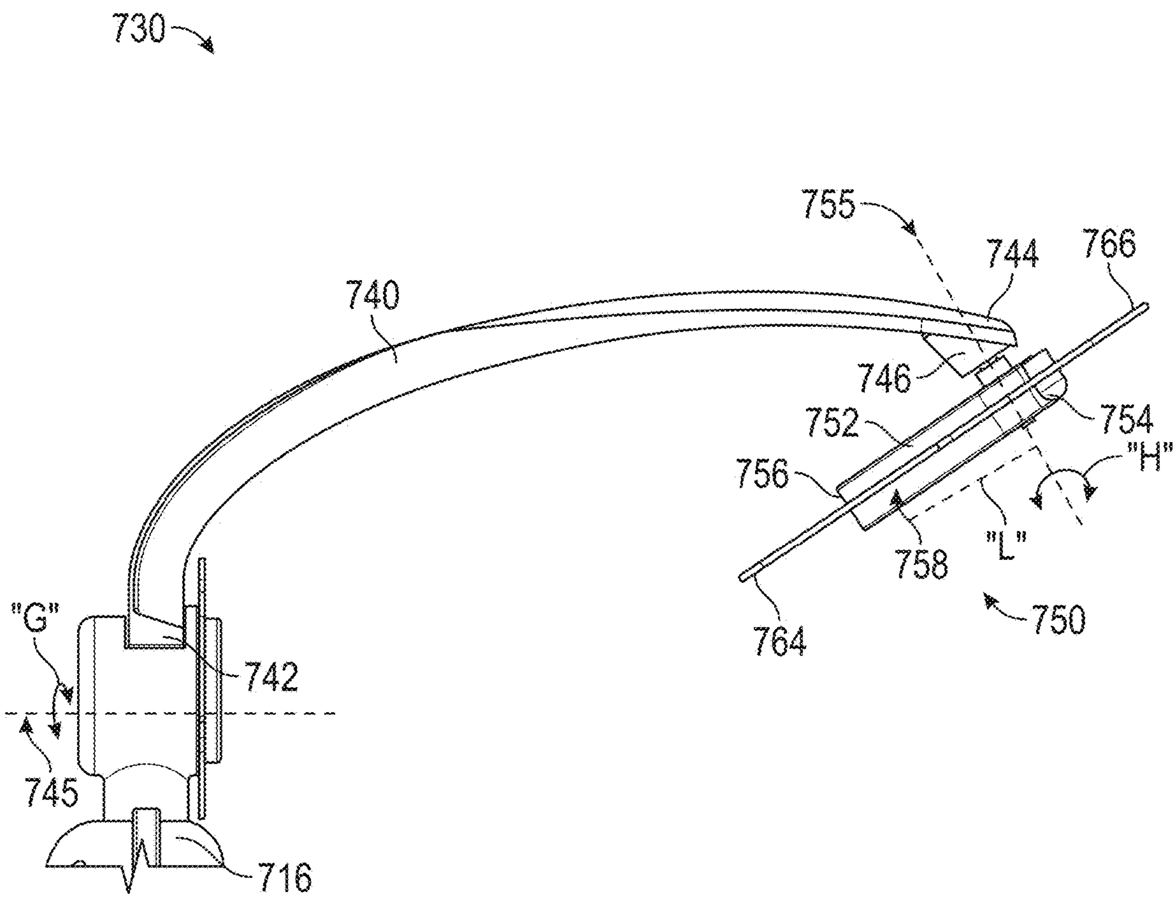

FIG. 15

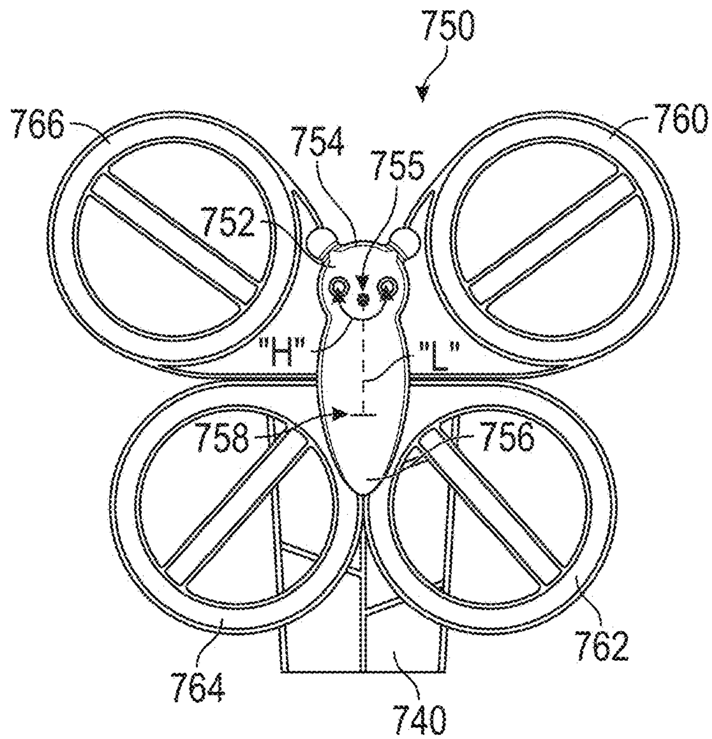

FIG. 16

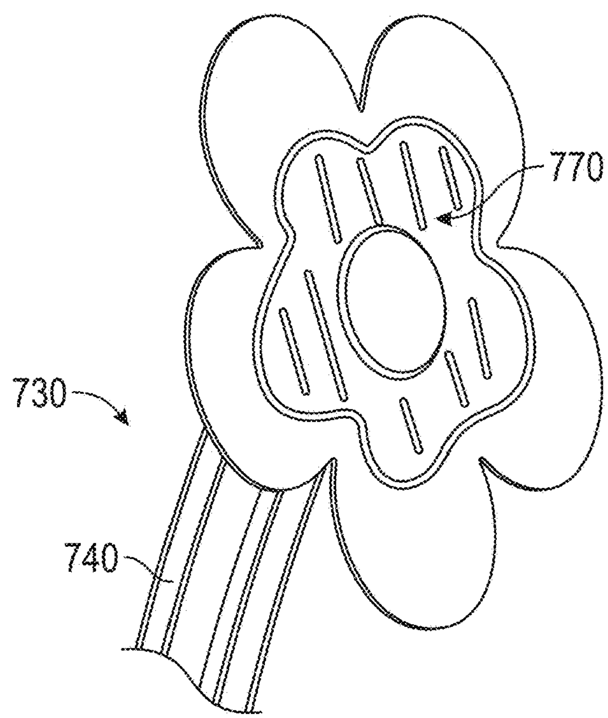

FIG. 17

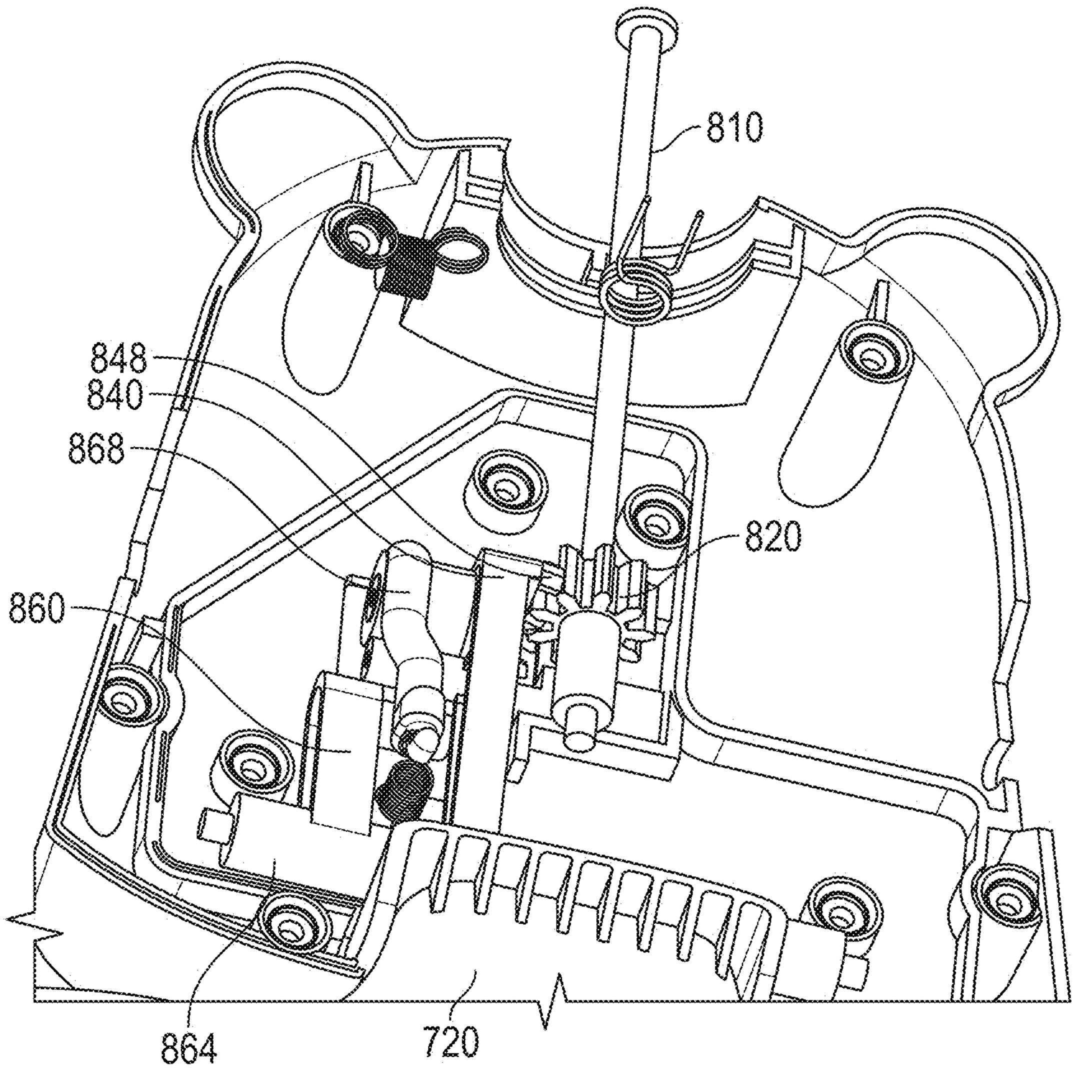

FIG. 18

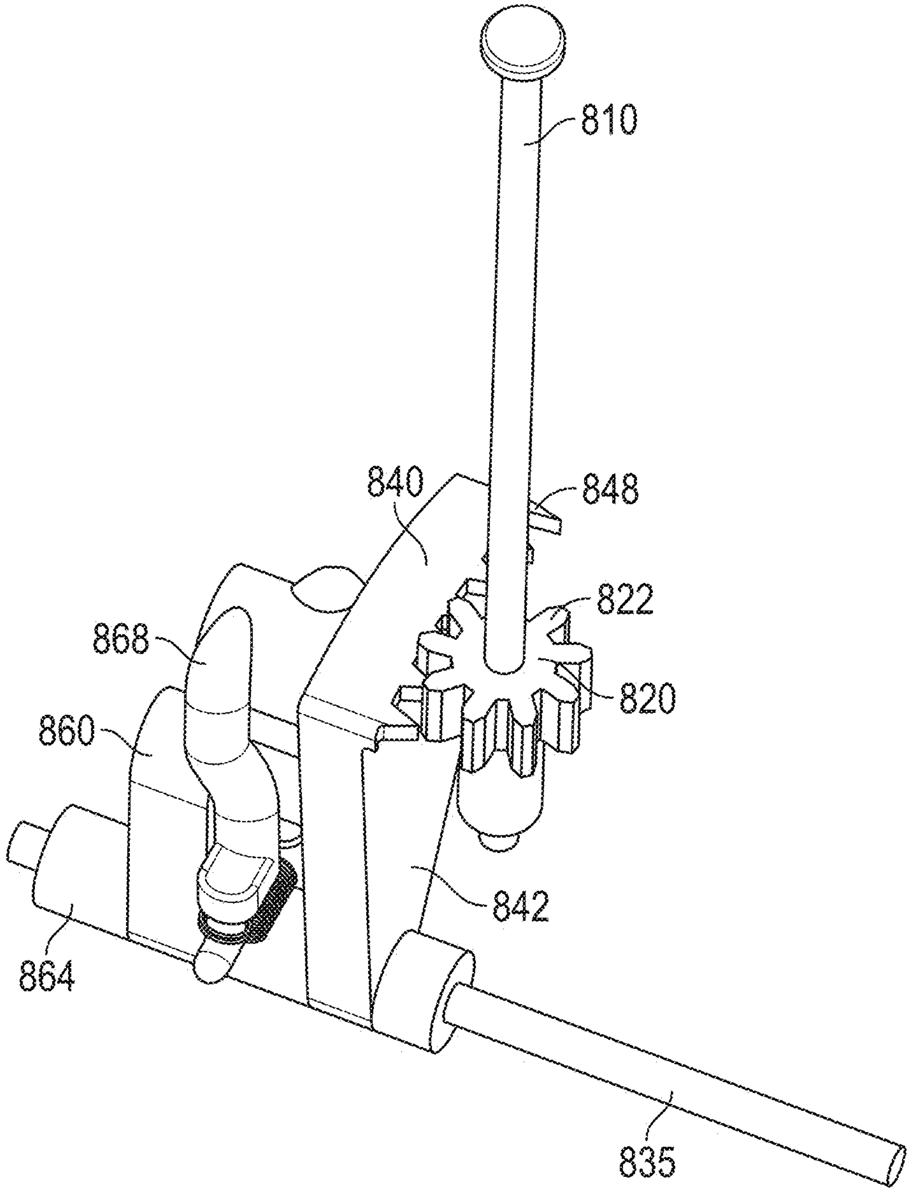

FIG. 19

INFANT ENTERTAINMENT DEVICE

TECHNICAL FIELD

The present disclosure relates to an infant entertainment device, and in particular, to an infant entertainment device with an entertainment portion that can be moved by an infant.

BACKGROUND OF THE INVENTION

Some conventional infant structures have mobiles that are static and not movable. Sometimes, such mobiles do not provide entertainment for an infant proximate to the conventional infant structures.

Therefore, there is a need for an infant entertainment device that has a portion that is engageable by an infant in a supine position. There is also a need for an infant entertainment device that has a portion that moves in response to infant inputs.

SUMMARY OF THE INVENTION

The present disclosure is directed to an infant entertainment device, comprising a support member for an infant, a housing positioned above the support member, an entertainment component rotatably coupled to the housing, the entertainment component being movable relative to the housing between a first position and second position, an input mechanism movably supported by the housing, the input mechanism being pivotally mounted and movable between a third position and a fourth position, the input mechanism being located above the support member so the input mechanism can be kicked by an infant lying on the support member, and a drive mechanism connected to the entertainment component and to the input mechanism, wherein movement of the input mechanism from its third position to its fourth position causes the drive mechanism to move the entertainment component from its first position to its second position.

In one embodiment, the drive mechanism includes a biasing member that engages the input mechanism, and the biasing member applies a force to the input mechanism so that the input mechanism moves from its fourth position to its third position.

In another embodiment, the biasing member is a first biasing member, the drive mechanism includes a second biasing member, and each of the first biasing member and the second biasing member engaging the input mechanism.

In an alternative embodiment, the housing includes an axle, the input mechanism is rotatably mounted on the axle, the input mechanism includes an engagement portion and a pair of mounting portions coupled to the engagement portion, each of the first biasing member and the second biasing member contacting one of the mounting portions, and each of the mounting portions being rotatably mounted on the axle.

In another embodiment, the entertainment component includes an arm and a moving component coupled to the arm at a pivot axis, the moving component being configurated to rotate about the pivot axis as the arm moves back and forth, the moving component including a body with a center of gravity that is offset from the pivot axis, and the movement of the body about the pivot axis is randomized by the center of gravity being offset from the pivot axis.

In one embodiment, the drive mechanism includes a first axle having a first gear mounted thereon, the first axle being connected to the entertainment component so that movement of the first gear results in rotation of the first axle and rotation of the entertainment component, and a second axle having a second gear mounted thereon, the second gear being engaged with the first gear so that rotation of the second gear drives the first gear.

In another embodiment, the drive mechanism includes a third axle having a clutch member rotatably mounted thereon, the clutch member being engaged with the second gear, the input mechanism is rotatably mounted on the third axle, and movement of the input mechanism drives the clutch member to rotate about the third axle which causes the second gear to rotate.

In yet another embodiment, the input mechanism includes a first keying member, the clutch member includes a second keying member, the second keying member being engaged with the first keying member so that movement of the input mechanism causes the clutch member to move.

In an alternative embodiment, the second gear includes a body having a first side and a second side opposite the first side, the body includes an arcuate toothed portion on the first side and a tab extending from the second side, the tab being engaged by the clutch member, and movement of the clutch member drives the tab of the body, thereby causing the second gear to rotate.

In one embodiment, the clutch member includes a clutch body having a mounting portion including the second keying member, the mounting portion being mounted on the third axle, and a drive portion coupled to the mounting portion, and a biasing mechanism, wherein the biasing mechanism is moved by the drive portion of the clutch body, the biasing mechanism engages the tab of the second gear, and movement of the biasing mechanism drives the tab and rotates the second gear.

In another aspect of the present invention, an infant entertainment device comprises a support member for an infant, a housing positioned above the support member, an entertainment component coupled to the housing and movable relative to the housing between a first position and second position, an input mechanism coupled to the housing and movable between a third position and a fourth position, the input mechanism being located above the support member so the input mechanism can be kicked by an infant lying on the support member, and a drive mechanism connected to the entertainment component and to the input mechanism, the drive mechanism includes a biasing member, wherein movement of the input mechanism from its third position to its fourth position causes the entertainment component to move from its first position to its second position, and the biasing member biases the input mechanism from its fourth position to its third position, thereby causing the entertainment component to return from its second position to its first position.

In one embodiment, the input mechanism is pivotally mounted to the housing and is located below the housing.

In another embodiment, the entertainment component is rotatably mounted to the housing and is located above the housing.

In yet another embodiment, the biasing member is a first biasing member, the drive mechanism includes a second biasing member, and each of the first biasing member and the second biasing member engages the input mechanism to bias the input mechanism from its fourth position to its third position relative to the housing.

In another embodiment, the drive mechanism includes a first axle having a first gear mounted thereon, the first axle being connected to the entertainment component so that movement of the first gear results in rotation of the first axle and rotation of the entertainment component, a second axle having a second gear mounted thereon, the second gear being engaged with the first gear so that rotation of the second gear drives the first gear, and a third axle having a clutch member rotatably mounted thereon, and the clutch member is engaged with the second gear.

In an alternative embodiment, the input mechanism is rotatably mounted on the third axle, and movement of the input mechanism drives the clutch member to rotate about the third axle which causes the second gear to rotate, the input mechanism includes a first keying member, the clutch member includes a second keying member, the second keying member being engaged with the first keying member so that movement of the input mechanism causes the clutch member to move.

In an alternative embodiment, the entertainment component includes an arm and a moving component coupled to the arm at a pivot axis, the moving component being configurated to rotate about the pivot axis as the arm moves back and forth, the moving component including a body with a center of gravity that is offset from the pivot axis, and the movement of the body about the pivot axis is randomized by the center of gravity being offset from the pivot axis.

In another aspect of the present invention, an infant entertainment device comprising a support member for an infant, a housing positioned above the support member, an entertainment component coupled to the housing and movable relative to the housing about a first axis between a first position and second position, an input mechanism coupled to the housing and movable about a second axis between a third position and a fourth position, the input mechanism being located above the support member so the input mechanism can be kicked by an infant lying on the support member, and a drive mechanism connected to the entertainment component and to the input mechanism, the drive mechanism includes a biasing member, wherein movement of the input mechanism about the second axis from its third position to its fourth position causes the entertainment component to move about the first axis from its first position to its second position, and the biasing member biases the input mechanism from its fourth position to its third position, thereby causing the entertainment component to return from its second position to its first position, the second axis being perpendicular to the first axis.

In an alternative embodiment, the input mechanism is pivotally mounted to the housing and is located below the housing, and the entertainment component is rotatably mounted to the housing and is located above the housing.

In another embodiment, the entertainment component includes an arm and a moving component coupled to the arm at a pivot axis, the moving component being configurated to rotate about the pivot axis as the arm moves back and forth, the moving component including a body with a center of gravity that is offset from the pivot axis, and the movement of the body about the pivot axis is randomized by the center of gravity being offset from the pivot axis.

Other systems, apparatuses, methods, features, and advantages will be, or will become, apparent to one with skill in the art upon examination of the following figures and detailed description. All such additional systems, apparatuses, methods, features, and advantages are included within this description, are within the scope of the claimed subject matter.

BRIEF DESCRIPTION OF THE DRAWINGS

The infant entertainment device disclosed herein may be better understood with reference to the following drawings and description. It should be understood that some elements in the figures may not necessarily be to scale and that emphasis has been placed upon illustrating the principles disclosed herein. In the figures, like-referenced numerals designate corresponding parts throughout the different views.

FIG. **1** is a schematic diagram of an embodiment of an infant entertainment device according to the present invention.

FIG. **2** is a perspective view of an embodiment of an infant entertainment device in a first configuration according to the present invention.

FIG. **3** is a perspective view of the infant entertainment device illustrated in FIG. **2** in a second configuration according to the present invention.

FIG. **4** is a front view of a portion of the infant entertainment device illustrated in FIG. **2** in a third configuration according to the present invention.

FIG. **5** is a perspective view of some components of another embodiment of infant entertainment device according to the present invention.

FIG. **6** is a front view of the components illustrated in FIG. **5**.

FIG. **7** is a front view of some components of another embodiment of infant entertainment device according to the present invention.

FIG. **8** is a front perspective view of the components illustrated in FIG. **7** with a portion of the housing removed.

FIG. **9** is a perspective view of a portion of the components illustrated in FIG. **7**.

FIG. **10** is a perspective view of some components of the drive mechanism of the infant entertainment device illustrated in FIG. **7**.

FIG. **11** is another perspective view of some components of the drive mechanism of the infant entertainment device illustrated in FIG. **7**.

FIG. **12** is a perspective view of one of the gears of the drive mechanism illustrated in FIG. **10**.

FIG. **13** is another perspective view of the gear illustrated in FIG. **12**.

FIG. **14** is a perspective view of another embodiment of an infant entertainment device according to the present invention.

FIG. **15** is a side view of a portion of the infant entertainment device illustrated in FIG. **14**.

FIG. **16** is a front view of an entertainment portion of the infant entertainment device illustrated in FIG. **14**.

FIG. **17** is a perspective view of some softgoods components of the infant entertainment device illustrated in FIG. **14**.

FIG. **18** is an internal view of some components of the infant entertainment device illustrated in FIG. **14**.

FIG. **19** is a top perspective view of some components of a drive mechanism of the infant entertainment device illustrated in FIG. **14**.

DETAILED DESCRIPTION OF THE INVENTION

An aspect of the infant entertainment device relates to an entertainment component that is moved by an input from an infant. In one embodiment, the entertainment component moves in response to a kicking input from an infant in a supine position. The infant entertainment device includes an input mechanism that is positioned so that the infant can move the input mechanism by kicking it.

The infant entertainment device includes a drive mechanism that converts movement of the input mechanism into motion of the entertainment component. In one implementation, the entertainment component is a mobile that moves when the infant kicks the input mechanism.

A schematic diagram of an embodiment of an infant entertainment device according to the present invention is illustrated in FIG. **1**. In this embodiment, the infant entertainment device **10** includes a pad **20** on which an infant can be positioned in an infant area **22**. In one use of the infant entertainment device **10**, an infant can be placed in a supine position on the pad **20**. The infant entertainment device **10** includes a housing or support housing **30** that is held in a position relative to the pad **20** by a support **32**. In one implementation, the support **32** is a support arm. In another implementation, there may be two support arms that hold the support housing **30**. The support housing **30** is positioned above the pad **20**.

Coupled to the support housing **30** is an engagement member **60** that can be moved relative to the support housing **30** between different positions. The engagement member **60** is coupled to the support housing **30** by one or more mounting arms **62**. In this embodiment, the engagement member **60** is mounted for movement relative to support housing **30** about axis **65**.

The engagement member **60** that can be engaged by an infant located in the infant area **22**. In particular, an infant in a supine position in the infant area **22** on the pad **20** can engage the engagement member **60** using the infant's feet. By kicking the engagement member **60**, the infant causes the engagement member **60** to move along the direction of arrow "A" about axis **65** from position **66** to position **68** relative to the support housing **30**. The support housing **30** includes a biasing member **35** that is engaged with the engagement member **60** as well. The biasing member **35** is positioned so that it applies a force on the engagement member **60** so that the engagement member **60** moves along the direction of arrow "B" about axis **65** from position **68** to position **66**. Thus, when an infant is no longer applying a force on the engagement member **60**, the biasing member **35** moves the engagement member **60** back to its initial position **66**. In different embodiments, the biasing member **35** can be a mechanism that can apply a force, such as a compression spring.

Infant entertainment device **10** includes an output member **40** that is movably supported by the housing **30**. The output member **40** can be referred to alternatively as an entertainment component. The output member **40** is positioned so that it is viewable by an infant located in the infant area **22**. A mounting arm **42** supports the output member **40** relative to the housing **30**. In one embodiment, an accessory **50** is coupled to the output member **40**. The accessory **50** can be fixedly or movably coupled to the output member **40**. In the implementation in which the accessory **50** is movably coupled to the output member **40**, movement of the output member **40** results in additional movement of the accessory **50** relative to the output member **40**, thereby increasing the visual stimulation for an infant on the pad **20**.

Infant entertainment device **10** includes a drive mechanism that is connected to the engagement member **60** and to the output member **40**. Different embodiments of drive mechanisms are described in detail below. Movement of the engagement member **60** causes the drive mechanism to move the output member **40**. In this embodiment, the output member **40** can be moved along the direction of arrow "C" about axis **45** from position **46** to position **48** when the engagement member **60** is moved from position **66** to position **68**. When the engagement member **60** moves from position **68** to position **66** due to biasing member **35**, the output member **40** moves along the direction of arrow "D" about axis **45** from position **48** to position **46**.

Turning to FIG. **2**, a perspective view of an embodiment of an infant entertainment device in a first configuration is illustrated. In this embodiment, the infant entertainment device **100** is shown in a first configuration **102**. Infant entertainment device **100** includes a support member or pad **110** that defines a support area or infant area **112** in which an infant can be placed. An infant can be placed in area **112** on pad **110** in a supine position or orientation.

Infant entertainment device **100** also includes a housing **120** that is connected to support arms **122** and **124** that are coupled to the pad **110**. The lower ends of the arms **122** and **124** can engage openings in the pad **110**, which holds the lower ends of the arms **122** and **124** in a fixed position relative to the pad **110**. Also coupled to housing **120** are a pair of arms **152** and **154** that support an engagement member **150**. The arms **152** and **154** are pivotally mounted relative to the housing **120**. When an infant (not shown) contacts an engagement surface **156** of the engagement member **150**, such as by kicking surface **156**, the engagement member **150** and arms **152** and **154** pivot relative to the housing **120**.

The housing **120** includes an upper portion or upper housing **126** to which an entertainment mechanism or entertainment component **130** is movably coupled. The entertainment component **130** is an overhead mobile. In this embodiment, entertainment component **130** includes several arms **132**, **134**, and **136** that have accessories **140**, **142**, and **144** coupled thereto. The entertainment component **130** can be rotated about axis **135** along the direction of arrow "C" and also rotated about axis **135** along the direction of arrow "D". When the engagement member **150** is moved by an infant, a drive mechanism connected to the engagement member **150** causes the entertainment component **130** to rotate about axis **135**.

Turning to FIGS. **3** and **4**, infant entertainment device **100** is illustrated in alternative configurations **104** and **106**, respectively. In configuration **104** (see FIG. **3**), arm **132** and accessory **140** have been decoupled from upper housing **126** and mounted into a base **133** located on one side of the pad **110**. In configuration **106** (see FIG. **4**), arm **132** and accessory **140** are the only items mounted to the upper housing **126**.

Turning to FIGS. **5** and **6**, perspective and front views of some components of another embodiment of infant entertainment device according to the present invention are illustrated. In this embodiment, the support housing of the infant entertainment device includes a gearbox housing **190** that defines a cavity **192**. Coupled to the gearbox housing **190** is an arm **132** that rotates along the direction of arrow "F" about an axis. The arm **132** can also be rotated in the opposite direction of arrow "F" as well.

Also, movably coupled to the gearbox housing **190** is an input mechanism **200**. In this embodiment, the input mechanism **200** includes an input member **220** that can be engaged by an infant, such as by the infant kicking the input member **220**. The input member **220** has an engagement member or engagement portion **230** that has an engagement surface **232**. Extending from the engagement portion **230** is a mounting portion **240** that is rotatably coupled to the gearbox housing **190**. In this embodiment, the mounting portion includes a pair of mounting arms **250** and **260** that define a gap or space **270** therebetween. Gearbox housing **190** extends into the gap **270** and includes an axle **280** extending therefrom.

Mounting arm **250** includes a sleeve **252** with a passageway extending therethrough. Similarly, mounting arm **260** includes a sleeve **262** with a passageway **264** extending therethrough. The axle **280** extends through passageways in the sleeves **252** and **262**. When the axle **280** rotates, the sleeves **252** and **262** and the mounting arms **250** and **260** rotate with the axle **280**. Thus, when an infant applies a force long the direction of arrow "E" on the engagement surface **232**, the input member **220** and the axle **280** rotate therewith.

Referring to FIG. **6**, gear **282** is mounted on axle **280**. As axle **280** rotates, gear **282** also rotates. The drive mechanism include an axle **290** that has a gear **292** mounted thereon. The teeth of gear **292** are engaged with the teeth of gear **282**. As a result, when gear **282** rotates, gear **292** and axle **290** also rotate. While in this example embodiment, gear **282** is a crown gear and gear **292** is a bevel gear, in other embodiments, the types of gears and their relative sizes can vary.

The drive mechanism also includes one or more biasing members that apply a force onto a rear surface of the input member **220** to move it in a direction opposite to arrow "E" about axle **280**. While there is a biasing member on the back side of each of mounting arms **250** and **260**, in FIG. **5** only biasing member **296** is shown. The engagement portion **230** is spring biased to return to its initial or default position by the biasing member **296**.

Turning to FIGS. **7-12**, various components of an alternative embodiment of an infant entertainment device according to the present invention are illustrated. Initially referring to FIGS. **7** and **8**, only part of the support housing, support arms, the drive mechanism, and the input mechanism are illustrated. The outer or front portion of the support housing has been removed to reveal the internal components. Infant entertainment device **300** includes a support housing **305** and an input mechanism **310** movably coupled to the support housing **305**. The input mechanism **310** includes an engagement portion **320** with an engagement surface **322**. The engagement surface **322** can be engaged by an infant, such as by kicking.

The input mechanism **310** is coupled to two arms **330** and **340** that are rotatably mounted relative to the support housing **305**. Arms **330** and **340** can be referred to alternatively as mounting portions. Arm **330** includes a sleeve **332** and a paddle portion **336** extending upwardly therefrom. Similarly, arm **340** includes a sleeve **342** and a paddle portion **346** extending upwardly therefrom. Sleeve **332** is located in a notch formed in the mount or flange **430** located in the support housing. Similarly, sleeve **342** is located in a notch formed in the mount or flange **440** located in the support housing.

In this embodiment, there are two biasing members **600** and **610** that engage the front surfaces of paddle portions **336** and **346** to bias the paddle portions **336** and **346** toward the rear of the support housing. In FIG. **7**, the front cover or front housing portion **410** of gearbox housing **400** is illustrated covering the gears of the drive mechanism. Turning to FIG. **8**, the front cover **410** has been removed, thereby revealing the various components of the drive mechanism **500** and the rear cover or rear housing portion **420**. The biasing members **600** and **610** and their relative positions to engage the paddle portions **336** and **346**, respectively, are shown in FIG. **8**. Also visible are the mounts or flanges **430** and **440**, and also the sleeves **332** and **342** of arms **330** and **340**.

In the gearbox housing **400**, axle **510** extends vertically and includes a gear **520** mounted thereon. In addition, axle **530** extends horizontally and includes a gear **540** mounted thereon. The drive mechanism **500** also includes an additional axle **535** that has a clutch **560** mounted thereon. As described in greater detail below, the clutch **560** is engaged with gear **540** and drives gear **540** to rotate about the axis defined by the longitudinal axis of axle **530**.

When an infant engages surface **322** of engagement portion **320**, movement of the engagement portion **320** moves arms **330** and **340** about the axis defined by the longitudinal axis of axle **535**. The sleeve **342** is engaged with the clutch **560** so that when sleeve **342** rotates, the clutch **560** rotates as well. As clutch **560** is engaged with gear **540**, the rotation of clutch **560** causes gear **540** and axle **530** to rotate. When gear **540** rotates, it causes gear **520** and axle **510** to rotate as well. Axle **510** is connected via to another gear that drives the entertainment member of the infant entertainment device.

Turning to FIG. **9**, a rear perspective view of a portion of some of the components illustrated in FIG. **7** is shown. In this view, the front housing portion **410** and the rear housing portion **420** that form the gearbox housing **400** are shown. The front housing portion **410** and rear housing portion **420** define an opening **435** therebetween. The clutch **560** extends through the opening **435** and includes a keying member **566**, which in this embodiment is a tab.

As shown, the sleeve **342** of the arm **340** includes keying member **344**, which in this embodiment is a notch. Keying member **344** receives keying member **566**, which results in the rotation of sleeve **342** causing clutch **560** to rotate as well. Extending from sleeve **342** is the paddle portion **346**.

FIGS. **10** and **11** are perspective views of some components of the drive mechanism **500** of the infant entertainment device, and FIGS. **12** and **13** are perspective views of gear **540**. As shown, gear **520** with teeth **522** is mounted to axle **510** and rotates therewith. Gear **540** is mounted to axle **530** and rotates therewith. Gear **540** includes a body **542** that has a first side **544** and an opposite second side **546** (see FIG. **13**). Extending from side **544** is a tab **550**. Extending from side **546** are teeth **548**, which are engaged with teeth **522** of gear **520**. In this embodiment, the teeth **548** are formed in an accurate shape. Also extending from side **546** is a sleeve **552** that has a passageway **554** through which axle **530** extends.

Clutch **560** includes a clutch body **562** with a mounting portion **564** that is mounted on axle **535**. The keying member **566** extends in a direction away from the mounting portion **564**. Extending from the mounting portion **564** is a drive portion **568** that is engaged with a biasing mechanism or biasing member **570**.

In this embodiment, the biasing mechanism **570** is a torsion spring that has a coiled portion **572** that has two extended ends **574** and **576**. End **574** extends through hole **569** formed in drive portion **568**. The gear **540** also includes a collar **556** (see FIG. **11**) that defines a receptacle for the biasing mechanism **570**. Located inside collar **556** is a shorter collar **558** that extends into the central opening of the coiled portion **572** of biasing mechanism **570**.

Referring to FIGS. **14-19**, an alternative embodiment of an infant entertainment device according to the present disclosure is illustrated. Referring to FIGS. **14-16**, a perspective view of infant entertainment device **700** is shown. Infant entertainment device **700** includes a base **710** from which two supports or support wires **712** and **714** extend to support a housing **716** therebetween. Movably coupled to the housing **716** is an engagement portion **720** that is positioned so that it can be engaged by an infant located proximate to the infant entertainment device **700**. When an infant engages the engagement portion **720**, such as by kicking engagement portion **720** with the infant's feet, the engagement portion **720** causes a drive mechanism of the infant entertainment device **700** to move an entertainment portion **730** relative to the housing **716**.

In this embodiment, the entertainment portion **730** includes a pivotally mounted arm **740** that reciprocates relative to the housing **716**. As shown in FIG. **14**, arm **740** has a driven end **742** and a free or distal end **744** that is opposite to the driven end **742**. Arm **740** is coupled to the housing **716**, and, in this embodiment, its driven end **742** is disposed in the housing **716**. The driven end **742** moves in a reciprocating or back and forth manner about pivot axis **745** along the directions of arrows "G". As driven end **742** moves back and forth about pivot axis **745**, the free end **744** of arm **740** swings back and forth about pivot axis **745** as well. In view of free end **744** being spaced apart from and at the opposite end of arm **740** from the driven end **742**, the distance that the free end **744** moves is much greater than the distance that the driven end **742** rotates.

As shown in FIGS. **14-16**, coupled to free end **744** of the pivotally mounted arm **740** is a moving component **750** which includes several display portions **760**, **762**, **764**, and **766** are coupled. In this embodiment, the free end **744** includes a housing **746** that defines a through hole that pivotally receives a mounting post extending from the moving component **750** to pivotally couple the moving component **750** to the arm **740**. The display portions **760**, **762**, **764**, and **766** are disposed so that they extend outwardly from the moving component **750** away from and relative to each other. Since the moving component **750** is coupled to the distal end **744** of the arm **740**, as the arm **740** swings back and forth, the moving component **750** will rotate relative to the arm **740**, thereby moving or spinning the display portions **760**, **762**, **764**, and **766** relative to the arm **740** and to the moving component **750**.

Referring to FIG. **16**, the moving component **750** includes a body **752** that has an upper portion **754** and a lower portion **756**. The body **752** is rotatably mounted to the distal end **744** of arm **740** and freely pivots about pivot axis **755** along the directions of arrows "H". In various embodiments, the weight of the body **752**, or the center of gravity of the body **752**, is offset from the pivot axis **755**. In this embodiment, the center of gravity is designated by **758**, which is shown in FIG. **16** by a distance "L" from the pivot axis **755**.

Referring back to FIGS. **14** and **15**, as the driven end **742** of arm **740** is moved back and forth about pivot axis **745** along the directions of arrow "G", the distal end **744** of the arm **740** rotates back and forth as well. The pivot connection between the distal end **744** and the moving component **750** allows the body **752** to rotate or swing back and forth about pivot axis **755** along the directions of arrow "H". As mentioned above, the distal end **744** of arm **740** moves a greater distance back and forth than the driven end **742** of arm **740** moves. The movement of the distal end **744** causes the body **752** to swing back and forth. Since the weight of the body **752** is offset from the pivot axis **755**, and depending its location, movement of the body **752** back and forth can sometimes result in the body **752** whipping around pivot axis **755** and making one or more full rotations about the pivot axis **755**. The movement of body **752** can be inconsistent and random based on the distance "L" that the weight or center of gravity of the body **752** is offset from pivot axis **755**.

The weight of the toy or the center of gravity **758** can be offset from the pivot axis **755** due to the shape and configuration of the body **752**, namely, the pivot axis **755** being located closer to the upper end of the body **752** than to the lower end of the body **752**. By varying the location of the center of gravity **758** of body **752**, and in particular, the distance "L", the movement of the body **752** can be adjusted. In alternative embodiments, a weight can be coupled to the body **752** at different locations, thereby changing the center of gravity **758** and the pattern of movement of the body **752**. Alternatively, a weight can be slidably mounted to the body **752** so its position relative to the pivot axis **755** is adjustable.

Turning to FIG. **17**, a perspective view of an embodiment of the entertainment portion **730** is illustrated. As shown, a softgoods covering **770** is coupled to the display portions **760**, **762**, **764**, and **766** (not shown in FIG. **17**) which provide support for the softgoods covering **770**.

Turning to FIGS. **18** and **19**, some of the components of the drive mechanism of the infant entertainment device **700** are illustrated. The drive mechanism includes an axle **810** that is coupled to the arm **740**. Rotation of axle **810** causes the arm **740** to rotate. Mounted on the axle **810** for rotation therewith is a gear **820** that has teeth **822** around its periphery. The drive mechanism includes a gear **840** that has a body **842** with teeth **848** arranged along an arcuate portion of the body **842**. As shown in FIG. **19**, gear **840** is mounted to axle **835** can rotate about axle **835**. The drive mechanism includes a clutch **860** with a mounting portion **864** and a drive portion **868**. When an infant pushes on the engagement portion **720**, the mounting portion **864** rotates with axle **835**. As a result, rotation of mounting portion **864** causes movement of the drive portion **868**, which in turn moves gear **840** via a biasing mechanism that engages both the drive portion **868** and the gear **840**.

In the foregoing detailed description, reference is made to the accompanying figures which form a part hereof wherein like numerals designate like parts throughout, and in which is shown, by way of illustration, embodiments that may be practiced. It is to be understood that other embodiments may be utilized, and structural or logical changes may be made without departing from the scope of the present disclosure. Therefore, the foregoing detailed description is not to be taken in a limiting sense, and the scope of embodiments is defined by the appended claims and their equivalents.

Aspects of the disclosure are disclosed in the description herein. Alternate embodiments of the present disclosure and their equivalents may be devised without parting from the spirit or scope of the present disclosure. It should be noted that any discussion herein regarding "one embodiment", "an embodiment", "an exemplary embodiment", or a similar phrase indicate that the embodiment described may include a particular feature, structure, or characteristic, and that such particular feature, structure, or characteristic may not necessarily be included in every embodiment. In addition, references to the foregoing do not necessarily comprise a reference to the same embodiment. Finally, irrespective of whether it is explicitly described, one of ordinary skill in the art would readily appreciate that each of the particular features, structures, or characteristics of the given embodiments may be utilized in connection or combination with those of any other embodiment discussed herein.

For the purposes of the present disclosure, the phrase "A and/or B" means (A), (B), or (A and B). For the purposes of the present disclosure, the phrase "A, B, and/or C" means (A), (B), (C), (A and B), (A and C), (B and C), or (A, B and C).

While the apparatuses and methods presented herein have been illustrated and described in detail and with reference to specific embodiments thereof, it is nevertheless not intended to be limited to the details shown, since it will be apparent that various modifications and structural changes may be made therein without departing from the scope of the inventions and within the scope and range of equivalents of the claims. For example, the infant positioner or infant support structures/apparatuses presented herein may be modified to contain any number of upstanding frame members, seat supports, interactive assemblies, interactive components, interactive elements, etc. Moreover, the infant positioner or infant support structures/apparatuses presented herein may be modified to resemble any other structure, device, etc.

In addition, various features from one of the embodiments may be incorporated into another of the embodiments. That is, it is believed that the disclosure set forth above may encompass multiple distinct inventions with independent utility. While each of these inventions has been disclosed in a preferred form, the specific embodiments thereof as disclosed and illustrated herein are not to be considered in a limiting sense as numerous variations are possible. The subject matter of the inventions includes all novel and non-obvious combinations and subcombinations of the various elements, features, functions, and/or properties disclosed herein. Accordingly, it is appropriate that the appended claims be construed broadly and in a manner consistent with the scope of the disclosure as set forth in the following claims.

It is also to be understood that terms such as "left," "right," "top," "bottom," "front," "rear," "side," "height," "length," "width," "upper," "lower," "interior," "exterior," "inner," "outer" and the like as may be used herein, merely describe points of reference and do not limit the present invention to any particular orientation or configuration. Further, the term "exemplary" is used herein to describe an example or illustration. Any embodiment described herein as exemplary is not to be construed as a preferred or advantageous embodiment, but rather as one example or illustration of a possible embodiment of the invention. Additionally, it is also to be understood that the infant positioner or infant support structures/apparatuses described herein, and any portions thereof, may be fabricated from any suitable material or combination of materials, such as plastic, metals, composites, etc., as well as derivatives thereof, and combinations thereof.

The terms "comprising," "including," "having," and the like, as used with respect to embodiments of the present disclosure, are synonymous. When used herein, the term "comprises" and its derivations (such as "comprising", etc.) should not be understood in an excluding sense, that is, these terms should not be interpreted as excluding the possibility that what is described and defined may include further elements, steps, etc. Similarly, where any description recites "a" or "a first" element or the equivalent thereof, such disclosure should be understood to include incorporation of one or more such elements, neither requiring nor excluding two or more such elements. Meanwhile, when used herein, the term "approximately" and terms of its family (such as "approximate", etc.) should be understood as indicating values very near to those which accompany the aforementioned term. That is to say, a deviation within reasonable limits from an exact value should be accepted, because a skilled person in the art will understand that such a deviation from the values indicated is inevitable due to measurement inaccuracies, etc. The same applies to the terms "about", "around", "generally", and "substantially."

What is claimed is:

1. An infant entertainment device, comprising:

a support member for an infant;

a housing positioned above the support member;

an entertainment component rotatably coupled to the housing, the entertainment component being movable relative to the housing between a first position and second position, the entertainment component including an arm and a moving component coupled to the arm at a pivot axis, the moving component being rotatable about the pivot axis as the arm moves back and forth, the moving component including a body with a center of gravity that is offset from the pivot axis, and movement of the body about the pivot axis is randomized by the center of gravity being offset from the pivot axis;

an input mechanism movably supported by the housing, the input mechanism being pivotally mounted and movable between a third position and a fourth position, the input mechanism being located above the support member so the input mechanism can be kicked by an infant lying on the support member; and

a drive mechanism connected to the entertainment component and to the input mechanism, wherein movement of the input mechanism from its third position to its fourth position causes the drive mechanism to move the entertainment component from its first position to its second position.

2. The infant entertainment device of claim **1**, wherein the drive mechanism includes a biasing member that engages the input mechanism, and the biasing member applies a force to the input mechanism so that the input mechanism moves from its fourth position to its third position.

3. The infant entertainment device of claim **2**, wherein the biasing member is a first biasing member, the drive mechanism includes a second biasing member, and each of the first biasing member and the second biasing member engaging the input mechanism.

4. The infant entertainment device of claim **3**, wherein the housing includes an axle, the input mechanism is rotatably mounted on the axle, the input mechanism includes an engagement portion and a pair of mounting portions coupled to the engagement portion, each of the first biasing member and the second biasing member contacting one of the mounting portions, and each of the mounting portions being rotatably mounted on the axle.

5. The infant entertainment device of claim **1**, wherein the drive mechanism includes:

a first axle having a first gear mounted thereon, the first axle being connected to the entertainment component so that movement of the first gear results in rotation of the first axle and rotation of the entertainment component; and

a second axle having a second gear mounted thereon, the second gear being engaged with the first gear so that rotation of the second gear drives the first gear.

6. The infant entertainment device of claim **5**, wherein the drive mechanism includes:

a third axle having a clutch member rotatably mounted thereon, the clutch member being engaged with the second gear, the input mechanism is rotatably mounted on the third axle, and movement of the input mechanism drives the clutch member to rotate about the third axle which causes the second gear to rotate.

7. The infant entertainment device of claim **6**, wherein the input mechanism includes a first keying member, the clutch member includes a second keying member, the second keying member being engaged with the first keying member so that movement of the input mechanism causes the clutch member to move.

8. The infant entertainment device of claim 7, wherein the second gear includes a body having a first side and a second side opposite the first side, the body includes an arcuate toothed portion on the first side and a tab extending from the second side, the tab being engaged by the clutch member, and movement of the clutch member drives the tab of the body, thereby causing the second gear to rotate.

9. The infant entertainment device of claim 8, wherein the clutch member includes:
a clutch body having:
a mounting portion including the second keying member, the mounting portion being mounted on the third axle, and
a drive portion coupled to the mounting portion; and
a biasing mechanism, wherein the biasing mechanism is moved by the drive portion of the clutch body, the biasing mechanism engages the tab of the second gear, and movement of the biasing mechanism drives the tab and rotates the second gear.

10. An infant entertainment device, comprising:
a support member for an infant;
a housing positioned above the support member;
an entertainment component coupled to the housing and movable relative to the housing between a first position and second position, the entertainment component including an arm and a moving component coupled to the arm at a pivot axis, the moving component being rotatable about the pivot axis as the arm moves back and forth, the moving component including a body with a center of gravity that is offset from the pivot axis, and the movement of the body about the pivot axis is randomized by the center of gravity being offset from the pivot axis;
an input mechanism coupled to the housing and movable between a third position and a fourth position, the input mechanism being located above the support member so the input mechanism can be kicked by an infant lying on the support member; and
a drive mechanism connected to the entertainment component and to the input mechanism, the drive mechanism includes a biasing member, wherein movement of the input mechanism from its third position to its fourth position causes the entertainment component to move from its first position to its second position, and the biasing member biases the input mechanism from its fourth position to its third position, thereby causing the entertainment component to return from its second position to its first position.

11. The infant entertainment device of claim 10, wherein the input mechanism is pivotally mounted to the housing and is located below the housing.

12. The infant entertainment device of claim 11, wherein the entertainment component is rotatably mounted to the housing and is located above the housing.

13. The infant entertainment device of claim 10, wherein the biasing member is a first biasing member, the drive mechanism includes a second biasing member, and each of the first biasing member and the second biasing member engages the input mechanism to bias the input mechanism from its fourth position to its third position relative to the housing.

14. The infant entertainment device of claim 10, wherein the drive mechanism includes:
a first axle having a first gear mounted thereon, the first axle being connected to the entertainment component so that movement of the first gear results in rotation of the first axle and rotation of the entertainment component;
a second axle having a second gear mounted thereon, the second gear being engaged with the first gear so that rotation of the second gear drives the first gear; and
a third axle having a clutch member rotatably mounted thereon, and the clutch member is engaged with the second gear.

15. The infant entertainment device of claim 14, wherein the input mechanism is rotatably mounted on the third axle, movement of the input mechanism drives the clutch member to rotate about the third axle which causes the second gear to rotate, the input mechanism includes a first keying member, the clutch member includes a second keying member, and the second keying member being engaged with the first keying member so that movement of the input mechanism causes the clutch member to move.

16. An infant entertainment device, comprising:
a support member for an infant;
a housing positioned above the support member;
an entertainment component coupled to the housing and movable relative to the housing about a first axis between a first position and second position, the entertainment component including an arm and a moving component coupled to the arm at a pivot axis, the moving component being rotatable about the pivot axis as the arm moves back and forth, the moving component including a body with a center of gravity that is offset from the pivot axis, and the movement of the body about the pivot axis is randomized by the center of gravity being offset from the pivot axis;
an input mechanism coupled to the housing and movable about a second axis between a third position and a fourth position, the input mechanism being located above the support member so the input mechanism can be kicked by an infant lying on the support member; and
a drive mechanism connected to the entertainment component and to the input mechanism, the drive mechanism includes a biasing member, wherein movement of the input mechanism about the second axis from its third position to its fourth position causes the entertainment component to move about the first axis from its first position to its second position, and the biasing member biases the input mechanism from its fourth position to its third position, thereby causing the entertainment component to return from its second position to its first position, the second axis being perpendicular to the first axis.

17. The infant entertainment device of claim 16, wherein the input mechanism is pivotally mounted to the housing and is located below the housing, and the entertainment component is rotatably mounted to the housing and is located above the housing.

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